Jiayi Ding, PE, PMP



Jiayi Ding is a senior bridge structural engineer with 12 years of experience in design, assessment, construction, and maintenance of transportation structures. He specializes in designing conventional and complex bridge structures including prestressed concrete, post-tensioned segmental, steel plate and tub girders, bridge substructure, deep foundations, and he is also experienced in high-fidelity finite element analysis, vessel collision, seismic, fatigue, and soil-structure interaction. Jiayi is proficient in multiple design standards including AASHTO LRFD, ACI, PCI, and various state DOT guidelines. He is a recognized contributor to the field, having presented at several prestigious conferences, published in peer-reviewed journals, and served as a reviewer for the ASCE Journal of Bridge Engineering. He also serves as an industry partner of CEE Department at University of Illinois. An ASCE member since 2013, Jiayi is a frequent speaker at ASCE events and an alumnus of the ASCE Younger Member Leadership Program of 2025.

Jiayi Ding's AtkinsRéalis project experience includes:

Central Florida Expressway Authority (CFX), SR 414 Expressway Extension (US 441 to SR 434) Project #414-208, Orlando, FL. Substructure Lead Engineer**.** AtkinsRéalis is Lead Designer (nearing 90%). Four bridges are included in the project, but the design effort centers around the 58-span elevated 4-lane viaduct (8638' total length) with 89' bridge width. Six spans will have four curved steel box girders with variable depth webs, varying 108" to 72" deep (from 9' at midspan to 6' at the supports) and spans up to 300' long. This is believed to be the first curved, steel box with variable depth webs designed in Florida. The remaining units are typical spans of six 78" deep prestressed concrete U-beams with spans up to 160'. The project is estimated to cost $500 million and the design schedule is 30 months. Four lanes of at-grade traffic will remain open during the day, with two lanes open at night in this 118' wide corridor. The viaduct will carry SR 414 express lanes over three existing bridges owned by FDOT in the median of the surface road. The single-column viaduct piers considered pseudo match-cast construction with post-tensioned "wings" connected to a cast-in-place column. The proposed inverted-T pier design lowers the profile by 10' through the residential community and represents a $60 million cost savings when compared to the PD&E segmental bridge concept. Jiayi led the substructure and foundation design of the 2 end bents and 56 piers, overseeing the design development and plan production of hammerhead and inverted T piers. Due to the unique geometry and loading carrying requirements, the two types of bridge substructure were designed and checked based on strut and tie method and refined finite element method in addition to conventional reinforced concrete design. He also led the design of deep foundation system utilizing steel H-piles and pipe piles that can withstand sinkhole extreme event loading and soil behavior. ***Duration: 11/2023 – 1/2026***

Florida Department of Transportation (FDOT) District 5, I-75 Phased Design Build Auxiliary Lanes from SR 44 to SR 200, Sumter and Marion Counties, FL. Quality Control Reviewer. The project involves three bridge replacements over I-75 (CR 462, CR 475 and SW 66th Ave) and the widening of the I-75 bridge over SR 44. AtkinsRéalis provided engineering services for structures, roadway, drainage, environmental/permitting, sign and pavement markings, ITS, utilities, landscaping

Total years of experience

12

Years with firm

2

Education

M.S., Structural Engineering, University of Illinois at Urbana Champaign, 2016

B.S., Civil Engineering, Michigan Tech, 2014

B.E., Civil and Structural Engineering, Hunan University, 2012

Registrations/licenses

Professional Engineer:
California 90653
Colorado 64996
Florida 88211
New York 1055285

Certifications

Project Management Professional (PMP), 2801313

Professional affiliations

American Society of Civil Engineers (ASCE)
Structural Engineering Institute (SEI)
American Institute of Steel Construction (AISC)
American Segmental Bridge Institute (ASBI)

Software

and aesthetics, and other roadway features. Jiayi performed Quality Control (QC) review for the design calculations, plans of bridge replacement (CR 462 over I-75, and SW 66th Ave over I-75). ***Duration: 11/2025 – 2/2026***

Federal Highway Administration (FHWA) Central Federal Lands Highway (FLHD), Big Oak Flat Road Tunnel Lighting Design, Yosemite Natl. Park, CA. Engineer of Record. This project involved structural design of utility conduit installations on three signature historic concrete arch bridges located within the park. Jiayi's responsibilities included designing the structural connections for suspending utility conduits beneath the bridge spans, ensuring compatibility with the existing historic structures while maintaining structural integrity. Jiayi also developed the technical specifications for the installation, GPR and damage remedy measures. ***Duration: 6/2024 – 2/2025***

Colorado Department of Transportation (CDOT) Region 1, I-70 (Floyd Hill to Veterans Memorial) Tunnels Design Engineering Services, Final Design & CDMG Services, Clear Creek County, CO. Engineer of Record for the Independent Design Check (IDC). Atkins is providing engineering services to improve travel-time reliability, increase highway capacity, replace a structurally deficient bridge, and enhance geometric design along westbound Interstate 70—from east of the Floyd Hill/Beaver Brook Exit (Exit 248) through the Veterans Memorial Tunnels to the Colorado Blvd./Idaho Springs Exit (Exit 241).The structural scope of the project includes the design of six bridges, including one major structure—Bridge A. Bridge A features a segmental cast-in-place post-tensioned concrete superstructure constructed using the balanced cantilever method. The segmental portion consists of four spans: two 280-foot side spans and two 347-foot intermediate spans. The superstructure is supported by post-tensioned hammerhead piers and architecturally distinctive twin-leaf slender piers, requiring advanced structural analysis and design to meet performance and aesthetic goals. Jiayi served as the Engineer of Record for the Independent Design Check (IDC) of the segmental portion of Bridge A along the Central Section of the project. This segmental bridge is a cast-in-place, post-tensioned concrete structure constructed using the balanced cantilever method. Jiayi led a team in the comprehensive verification of the segmental superstructure design for servuce and construction stages, as well as the review of substructure components including mildly reinforced twin-leaf piers, post-tensioned concrete hammerhead piers, and drilled shaft foundations. The unique twin-wall pier configuration required high-fidelity moment-curvature analysis to determine effective stiffness based on cracked section behavior—an approach not explicitly addressed in the AASHTO LRFD Bridge Design Specifications. Jiayi performed detailed load calculations and assessed moment amplification effects on substructures due to secondary load paths. He conducted cracked section analysis and evaluated tensile stresses in column reinforcement during both construction and service stages. ***Duration: 12/2023 – 5/2025***

Georgia Department of Transportation (GDOT), SR 4 BU/US 1 BU/ US 23 BU at CSX Railroad Ware County, GA. Quality Control Reviewer. This project aims to construct a grade-separated bridge to eliminate delays caused by trains blocking the existing at-grade railroad crossings. The new bridge (15 spans, 1700 ft total length) will reduce travel times and minimize the risk of train-vehicle and train-pedestrian interactions along Ossie Davis Parkway. Additionally, it will provide a grade-separated crossing over an associated spur line. Several intermediate bents of this bridge are post-tensioned inverted T straddle bents, which requires refined modeling and analysis. Jiayi performed Quality Control (QC) review for the design of post-tensioned straddle inverted-T bents, ensuring accuracy and compliance with project standards. His responsibilities included verifying structural models in MIDAS and

ANSYS
AutoCAD
Bluebeam
CSI-Bridge
FB-MultiPier
LARSA
LEAP Bridge Suite
MathCAD
MDX
ProjectWise

Publications

"Design of Complex Bridge Substructure using Strut and Tie Method" Structures Congress, 2026

"Soil Structure Interaction Analysis of Balanced Cantilever Segmental Bridges" Structures Congress, 2026

"Damage Detection in FRP-Reinforced Concrete Elements" Materials, 2024

"A Novel Rehabilitation Solution for Precast Concrete Slab Beam Bridges" International Bridge Conference, 2024

"Diagnostic Load Testing of a Bridge Before and After Rehabilitation" Bridge Engineering Institute Conference, 2023

Presentations

"Design and Analysis of a Complex Post-Tensioned Segmental Bridge", ASCE Florida Section Annual Conference, July 2025

"Design of Bridge Inverted-T and Hammerhead Pier Based on Strut and Tie Method", ASCE Florida Section Annual Conference, July 2025

"Enhancing Bridge Infrastructure Health and Longevity", 1st Symposium on Climate-Smart Infrastructure Innovations & Implementation (CSI3), March 2025

"Design, Construction and Validation of a Rehabilitation Technique for PC Slab Bridges", Transportation Infrastructure Precast Innovation

reviewing calculations related to service stress checks, flexural moment capacity, shear and torsion demands, and corbel design. Jiayi also cross-checked design outputs against construction plans to ensure consistency and constructability. ***Duration: 9/2024-9/2025***

Experience Prior to Joining AtkinsRéalis

Broward County Department of Public Works, HBMD, Pompano Beach, FL. In-House Consultant and Project Manager. After the partial collapse of a FDOT Sonovoid bridge in 2017, Jiayi led innovative rehabilitation efforts as an in-house consultant and project manager. He developed a near-surface-mounted-dowel repair method, improving lateral load distribution by 25%. Jiayi managed asset evaluations, prioritized repairs, and coordinated with agencies, enhancing bridge performance and promoting the adoption of this method for structurally deficient bridges. ***Duration: 5/2021-10/2022***

Florida Department of Transportation (FDOT) District 4, Roosevelt Bridge Emergency Repair, Stuart, FL. Independent Design Review Engineer. During a routine biennial inspection period, FDOT inspectors found cracks on the southernmost span of the southbound Roosevelt Bridge. FDOT immediately assembled a statewide team of bridge experts who began assessing the situation and performing a full structural review. Jiayi's role involved conducting an independent review of the proposed retrofitting design on the damaged first unit, which consists of 5 balanced cantilever continuous spans, with span lengths vary from 196 ft to 260 ft. Jiayi performed the analysis of the structure, for evaluating the normal stresses, principal stresses, ultimate moment capacity, shear and torsion capacity to verify the effectiveness of the new external post-tensioning for replacing the compromised post-tensioning and restoring the girder capacities, and he also reviewed the details and constructability of the new tendon deviators proposed by the designer, and provided. several comments regarding bridge jacking and foundation evaluation. ***Duration: 6/2020-8/2020***

Florida Department of Transportation (FDOT) District 4, SR 710/Beeline Highway (Northlake Boulevard to Blue Heron Boulevard) Design Services, Palm Beach County, FL. Structural Engineer. Jiayi contributed to the SR 710 capacity improvements project, which involved widening from a 4-lane to a 6-lane urban facility. He designed multi-column-cap piers for a new bridge, reviewed PC/PS I girder designs, and performed bridge load rating analysis using LEAP Bridge software. Additionally, Jiayi designed mile-long cantilever retaining walls for the junction slab and sidewalk using FDOT Mathcad. ***Duration: 12/2018-4/2022***

Florida Department of Transportation (FDOT) District 4, SR A1A North Bridge over ICWW Design Services, St. Lucie County, FL. Structural Engineer. The main scope of this project is to replace the existing bascule bridge with a new high-level fixed bridge over the Intercoastal Waterway (ICWW), which will have a navigational clearance of 85 ft vertical and 125 ft horizontal at the waterway channel (estimated construction cost is $111.5 million). The new SR A1A Bridge will have a total length of 4152.5 ft, consist of 23 simple spans, and a 3-span continuous channel unit over the ICWW. The superstructure will use Florida I-beams. The two-column-cap piers will be supported on oval-shaped caps and driven piles. Jiayi assisted the EOR for leading the production of substructure and foundation design from 60% to final completion. He used FB-MultiPier software for analyzing the bridge's channel unit's behavior subject to vessel collision and scour, considering nonlinear behavior of soil-structure-interaction. He also performed AASHTO vessel collision Method II analyses on the other twelve water piers using the FDOT vessel collision analysis

Center (TRANS-IPIC) TIP Day, November 2024

"Rehabilitation Solutions for Precast Concrete Slab Beam Bridges", Transportation Research Board (TRB) AKB 30 Concrete Bridge Committee Mid-Year Meeting, August 2024

"Bridge Foundation Design", ASCE Florida Section Annual Conference, July 2024

Awards

SEI Young Professional Tier 1 Scholarship to Structures Congress 2026

MathCAD program. Working closely with the superstructure designer and geotechnical engineer, he completed the design of reinforced concrete pier caps, columns, pile caps and driven piles of the water piers. The design process included load calculation, flexural and shear design, moment amplification for secondary effect, reinforcement service stress, pile design and lateral stability analysis. ***Duration: 12/2019-12/2021***

Florida Department of Transportation (FDOT) District 5, I-4 Ultimate Design-Build, Orlando, FL. Structural Engineer. The main scope of this mega design-build project consists of design, build, maintain and operate I-4 through downtown Orlando. The design-build project, with an estimated design and construction cost of $2.3B. I-4 will be completely transformed along the 21 miles that stretch from Kirkman Road to SR 434. The project included reconstructing 15 major interchanges, adding 85 extra lane-miles of mainline roadway and constructing more than 145 bridges. The four bridges that Jiayi worked on were #750747 (steel I-girders, 2466 ft overall length, 16 spans, main span is 257 ft), #750750 (steel box girders, 647.5 ft overall length, 4 spans, main span is 191.75 ft with a 289 ft radius), #750751 (steel box girders, 3154.75 ft overall length, 16 spans, main span is 266.6 ft), and #750799 & #750790 twin bridges (steel I-girder, 607 ft overall length, main span is 307 ft). Jiayi provided post-design construction support services for four complex steel tub girder bridges, ensuring structural integrity and compliance with design specifications and regulatory standards. His responsibilities included detailed review of steel shop drawings, girder erection analysis, and load rating evaluations. Jiayi performed advanced shell element modeling using LARSA 4D to assess stress concentrations near openings in the steel tub girders. His analysis identified critical stress zones, leading to the design and implementation of strengthening measures at the steel box girder bearing soleplates to ensure uniform stress distribution and prevent overstressing of the end diaphragms. He also conducted deck pour sequence analysis and calculations using LARSA 4D, and developed custom load rating spreadsheets for the steel box girders. In addition, Jiayi supported bridge load rating updates, prepared bridge plan revision packages, and responded to contractor Requests for Information (RFIs) and Requests for Modification (RFMs). ***Duration: 6/2018-12/2020***

Florida Department of Transportation (FDOT) District 6, I-395 Signature Bridge, Miami-Dade County, FL. Structures Engineer. This large-scale design-build project includes the construction of a high-profile signature bridge across the Biscayne Boulevard. The signature bridge will have the largest precast concrete segmental arch in North America. The bridge's tallest arch will soar 325 feet above the ground and span 650 feet across. As a structural engineer on the Signature Bridge design team, Jiayi Ding was responsible for the detailed analysis and design of key arch components and supporting structures. His work included longitudinal design of Arches 1 and 4, addressing normal stress, principal stress, ultimate flexural capacity, shear, and combined shear-torsion effects. He also performed transverse analysis of precast post-tensioned arch segments under cable-induced loads to ensure structural performance and stability.Jiayi completed the analysis and design of cast-in-place reinforced concrete starter segments and designed the end bents, contributing to the overall integrity and constructability of the bridge system. ***Duration: 1/2019-12/2019***

Florida Department of Transportation (FDOT) District 6, Palmetto Expressway Express Lanes, Miami-Dade County, FL. Structural Engineer. The SR 826/Palmetto Expressway Capacity project aimed at widening the corridor for increased traffic volume and reducing congestion (estimated construction cost is $66 million). A critical component of this project is to modify the flyover ramp bridge to

accommodate the addition of a permanent general-purpose lane to the SR 826. The proposed bridge modification involves partial deck removal, extraction of exterior steel girders, partial demolition of hammerhead piers, and reconstruction of the existing c-pier (Pier 5), including the deck and traffic barriers. A comprehensive evaluation was undertaken to understand the bridge's behavior and load transfer dynamics during construction stages. A full construction stage analysis was employed to address the challenges posed by the special construction stages of the bridge modification and load transfer. Jiayi led a comprehensive feasibility study for the 103rd Street Ramp Bridge within a three-month timeframe, supporting the development of the Request for Proposal (RFP). His work included structural analysis of steel girders using MDX software to verify stress and shear demands during both construction stages and final service conditions.
Jiayi developed a global bridge model to perform detailed construction stage analysis, focusing on load transfer effects from the existing C-pier to the newly proposed C-pier. This analysis informed the evaluation of substructure and foundation demands under evolving load conditions.
He completed the deck reinforcement design and performed flexural and shear checks for the hammerhead piers, with particular attention to the continuity of top reinforcement in the pier caps to resist increased negative moments following partial demolition. Jiayi also prepared the conceptual layout and sizing of the new C-piers and conducted a thorough reassessment of the foundation system, accounting for residual effects from permanent loads and redistribution of live loads among the piles. ***Duration: 8/2020-12/2020***

Florida Department of Transportation (FDOT) District 6, SR 25/Okeechobee Road (East of 116th Way to East of NW 87th Avenue), Miami-Dade County, FL. Structural Engineer. This project involves the design and reconstruction of SR 25/Okeechobee Road from east of NW 116th Way to east of NW 87th Avenue to reduce congestion and allow uninterrupted flow for vehicles travelling on Okeechobee Road (estimate construction cost is $105.6 million). Part of this large-scale infrastructure improvement project is to reconstruct the NW 121 Way bridge through demolishing the existing deck and pouring a new widened reinforced concrete flat slab over the existing substructure and foundation. Additionally, the new NW 121 Way bridge features highly curved edges, necessitating advanced finite element analysis. This analysis was essential for accurately determining load demands along the curved portions of the flat slab. Jiayi developed a refined plate-based model and conducted vehicular analysis for the highly curved flat slab bridge. Through this process, he obtained the envelops of moment and shear demands within the curved exterior portions of the bridge, which exceeded the exterior slab demands obtained using the AASHTO simplified "strip" method. The refined bridge model also led to a more accurate representation of lateral load distribution and load path considering the bridge skew. Based on the load demands from analysis, he designed the orthogonal reinforcement in the flat slab. He also verified the existing substructure and foundation for sustaining the new loads. ***Duration: 9/2019-6/2021***

Fort Lauderdale-Hollywood International Airport Terminal 5, Fort Lauderdale, FL. Structural Lead Engineer. The Fort Lauderdale-Hollywood International Airport will be expanding with Terminal 5, a new concourse with added gates. The new 230,000-square-foot terminal will be constructed at the airport’s east end. The project will use construction manager at risk (CMAR) delivery method. A part of the Terminal 5 project was to design and construct a flyover bridge to carry a single lane of traffic exiting from the new terminal 5 to the Terminal Drive. The proposed flyover bridge has 4-span continuous steel plate girders, incorporating two 99 ft side spans, two 165 ft interior spans, with a total length spanning 528 ft. The bridge faced notable design challenges due to its geometric constraints, including the 160 ft

radius and shallow girder depth to meet vertical clearance. The geometric constraints led to design features that require refined analysis and specific treatment. Jiayi served as the lead designer for a uniquely curved, four-span continuous steel tub girder bridge with a tight horizontal radius of 160 feet. Jiayi was responsible for advancing the project from conceptual design through final design, demonstrating expertise in complex geometry and structural behavior.
He prepared the Bridge Development Report (BDR), evaluating both steel I-girder and box girder alternatives. His preliminary analysis included ultimate strength and service load combinations, leading to the selection and detailed design of the steel I-girder system. Jiayi also designed the hammerhead piers, end bents, and deep foundations utilizing 24-inch driven piles. Given the bridge's sharp curvature, Jiayi conducted a parametric study using high-fidelity finite element modeling with shell elements in CSI Bridge. This analysis accounted for the significant torsional effects and the role of intermediate cross-frames in overall bridge behavior. To mitigate uplift forces at the bearings on the end bents, he designed reinforced concrete end diaphragms, ensuring structural stability and serviceability. ***Duration: 9/2022-9/2023***

New York City Department of Transportation (NYDOT), Queensboro Bridge Upper Roadway Deck Replacement, Queens, NY. Structural Engineer. The Queensboro bridge opened to traffic on June 18, 1909, as the longest cantilever bridge in the United States. This five-span cantilever truss bridge, designed to accommodate heavier loads, is the only one of the four great East River Bridges that is not a suspension bridge. This project involves providing design and construction support for the reconstruction of the upper deck on main spans. With 200,000 square feet of roadway deck replacement needed due to years of overloading, the challenge lies in selecting a lightweight, durable replacement for extended service life. To address this, a novel open-rib steel orthotropic deck with a polymer overlay was designed, maximizing stiffness-to-weight ratio for the long span cantilever truss bridge. Jiayi was responsible for the structural assessment and retrofit design of an open-rib orthotropic deck system, with a focus on fatigue performance under cyclic loading conditions. His work included evaluating maximum von Mises stresses through detailed fatigue analysis and developing retrofit solutions for critical components such as floor beams, sub-floor beams, support diaphragms, and their connections. Jiayi performed nonlinear finite element fatigue analysis using ANSYS Mechanical to capture complex stress behavior and fatigue life predictions. In addition, he designed an innovative, customized steel traffic railing system that met AASHTO TL-5 crash testing requirements, ensuring both safety and compliance with national standards. ***Duration: 6/2016-10/2017***

New York Metropolitan Transportation Authority (MTA), Seismic Retrofit of Approaches at Throgs Neck Bridge, Bronx, NY. Structural Engineer. Jiayi served as the main investigator for the Throgs Neck Bridge Approach Viaducts Rehabilitation project, focusing on seismic vulnerability and rehabilitation design to extend the viaducts' service life by 25 years. He reviewed past reports, calculated the characteristics parameters of seismic isolation bearings, and conducted comprehensive modeling and analysis, including nonlinear time-history and response spectrum analyses. Jiayi also co-authored the technical report and coordinated project tasks. ***Duration: 9/2016-10/2017***

New York Metropolitan Transportation Authority (MTA), Throgs Neck Bridge Suspended Span Deck Replacement, Bronx, NY. Structural Engineer. The scope of this project was to design the replacement of the deck system on the suspended spans of the Throgs Neck Bridge. The existing deck system, a concrete-filled grid with relief joints at every sixth-floor truss, was approaching the end of its useful service life. In addition, the relief joints are leaking, causing corrosion on the

stringers, floor trusses, and lateral bracing below. Thus, a new closed-rib orthotropic deck system was designed. A preliminary wind study and analysis were conducted to check the bridge stability against a 10,000-year return period. The predicted mean hourly wind speeds at the deck level of the two joint-less deck alternatives far exceeded the 10,000-year wind speed (115 mph), meaning the bridge is very stable against strong winds. Wind tunnel testing will also be performed for the proposed structure, which will include the impacts of staging and wind on trucks. Jiayi was responsible for the structural analysis and design of a steel closed-rib orthotropic deck system subjected to fatigue truck loading. His work included evaluating fatigue performance and stress behavior under repetitive loading conditions, ensuring long-term durability and compliance with design standards. He also designed strengthening measures for floor beams and their connections to enhance structural capacity and service life. Additionally, Jiayi performed nonlinear analysis and developed a customized steel traffic railing system that meets AASHTO TL-5 crash testing requirements, combining safety, innovation, and structural efficiency. ***Duration: 6/2016-10/2017***

Ontario Line TA Planning, Metrolink, Thorncliffe Segment, Toronto, Canada. Structural Lead Engineer. The Don Valley Crossing is five-span balanced-cantilever segmental bridge spanning a total length of 505 meters (1657 ft) with the main span measuring 138 meters (453 ft). The reinforced concrete substructure consists of three twin-wall piers and two single-column piers to mitigate the creep and shrinkage effects stemming from an unbalanced span layout and various construction stages, the bridge design involved refined analysis based on substructure's cracked section stiffness. This type of refined analysis considering the flexible twin-wall piers in accommodating creep, shrinkage and negative unform temperature is not addressed in AASHTO publications. Jiayi led a structural design team through a series of advanced analyses to support the design and evaluation of segmental bridge substructures. The unique twin-wall piers requires high fidelity moment curvature analysis to determine the effective stiffness based on cracked section analysis which is not commonly addressed in AASHTO code. His responsibilities included performing load calculations and assessing moment amplification effects on substructures due to secondary load paths. He conducted cracked section analysis of columns and evaluated tensile stresses in column reinforcement during both construction and service stages. Utilizing Xtract software, Jiayi performed detailed nonlinear section analysis to ensure accurate modeling of structural behavior and reinforcement performance under realistic loading conditions. His work contributed to the development of robust and reliable substructure designs that meet project requirements and industry standards. ***Duration: 9/2022-9/2023***

Research Triangle Regional Public Transportation Authority (GoTriangle), Durham, NC. Structures Designer. The D-O Light Rail Transit project will span 17.7 miles, offering more than 26,000 daily trips to Durham and Chapel Hill residents and commuters. The proposed light rail bridge will feature a superstructure composed of precast segments, constructed through balanced cantilever or span-by-span methods. The final design of GoTriangle's 17.7-mile Durham-Orange light rail system, includes design of a 450-foot, 3-span balanced-cantilever unit and three 125-foot simple-span units. Jiayi performed longitudinal post-tensioned tendon design of one three-span balanced-cantilever unit (450 feet) using internal post-tensioned tendons and three span-by-span units (125 feet) using external post-tensioned tendons and deviators. The longitudinal analyses included service load check (normal stresses, principal stresses), ultimate load check (shear and torsion, ultimate moment capacity), and construction stresses check. Developed unit models on LARSA 4D and performed construction stage analysis and transit load analysis. Designed longitudinal (torsion) reinforcement, transverse (shear, torsion)

reinforcement, temporary post-tensioned bars for cast-in-place wet joints of a balanced cantilever unit and checked the design of span-by-span unit diaphragm based on strut-and-tie method. ***Duration: 10/2018-3/2019***